80Y

80Y

12K

80Y

12K

12.5B

80Y

12K

12.5B

3,500 KM

Motion
Consulting

Prepared for

Prairie Oil

Diya Arora

Kritin Kumar

Daisy Dyck

Christian Arteaga

Motion Consulting

How can Prairie Oil prepare for growing cybersecurity threats in the industry?

Executive Summary

How can we help?

Motion Consulting

Problem Statement

How can Prairie Oil prepare for growing cybersecurity threats in the industry?

Criteria

- Change Management
- IT & OT Compatibility
- Cybersecurity
- Financial Feasibility

Solution

Keep It Moving

Impacts

- 4 Billion Dollars
- 24 Hours
- Zero Incidents

Personal Impacts

How does this affect the individuals?

Motion Consulting

Vivek

Age: 36 (IT Lead)

- Has worked with Prairie Oil for 5 years now
- Dreads audits asking him, "*How is your system secure?*"
- Wants a system that allows patching without risking downtime
- Struggles to balance security with day-to-day IT duties

Personal Impacts

How does this affect the individuals?

Vivek

Age: 36 (IT Lead)

- Has worked with Prairie Oil for 5 years now
- Dreads audits asking him, "*How is your system secure?*"
- Wants a system that allows patching without risking downtime
- Struggles to balance security with day-to-day IT duties

Kaitlynn

Age: 46 (Pipeline Supervisor)

- Has worked at Prairie Oil for 17 years
- Pipeline supervisor for a large sector of pipelines across Alberta
- "If it isn't broken, don't fix it"

Pain Point:

- Still relies on the legacy Honeywell TDC 3000, but fears change

People, Process, Technology

Motion Consulting

Current State

- Stressed by change
- Facing nuisance due to legacy systems
- Unprepared for cyber threats

Desired State

People, Process, Technology

Motion Consulting

Current State

- Stressed by change
- Facing nuisance due to legacy systems
- Unprepared for cyber threats

Desired State

- Prepared for growing cybersecurity risks
- Reduced workload due to outdated systems

People, Process, Technology

Motion Consulting

Current State

- Stressed by change
- Facing nuisance due to legacy systems
- Unprepared for cyber threats

Key Enabler

Updated operations with a change management implementation plan

Desired State

- Prepared for growing cybersecurity risks
- Reduced workload due to outdated systems

People, Process, Technology

Motion Consulting

Current State

Desired State

Key Takeaway

Employees will be vulnerable in the event of any system changes

Criteria:
Change Management

People, Process, Technology

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Current State

- 3 Canadian refineries with siloed data, lacking security measures
- Data analyzed locally and sent to the head office via microwave
- IT and OT integration relies on ad-hoc solutions

Desired State

People, Process, Technology

Motion Consulting

Current State

- 3 Canadian refineries with siloed data, lacking security measures
- Data analyzed locally and sent to the head office via microwave
- IT and OT integration relies on ad-hoc solutions

Desired State

- Centralized data storage and management
- Independent Integration between IT and OT systems
- Create “perfect fit” rather than “functional”

People, Process, Technology

Motion Consulting

Current State

- 3 Canadian refineries with siloed data, lacking security measures
- Data analyzed locally and sent to the head office via microwave
- IT and OT integration relies on ad-hoc solutions

Key Enabler

Completing migration to new operating system and finding a compatible data storage system

Desired State

- Centralized data storage and management
- Independent Integration between IT and OT systems
- Create "perfect fit" rather than "functional"

People, Process, Technology

Motion Consulting

Key Takeaway

Integrated independence between IT & OT must be stronger

Criteria:
IT & OT Compatibility

Current State

- 3 Canadian refineries with siloed data, lacking security measures
- Data analyzed locally and sent to the head office via microwave
- IT and OT integration relies on ad-hoc solutions

Key Enabler

Completing migration to new operating system and finding a compatible data storage system

Desired State

- Centralized data storage and management
- Solid integration between IT and OT systems
- Create "perfect fit" rather than "functional"

People, Process, Technology

Motion Consulting

Current State

- Using Honeywell TDC 3000, which in 2025 will be 41 years old
- Partial migration to Siemens PCS 7, leaving fragmented security practices

Desired State

Analysis Recommendation Implementation Impacts Risks

People, Process, Technology

Motion Consulting

Current State

- Using Honeywell TDC 3000, which in 2025 will be 41 years old
- Partial migration to Siemens PCS 7, leaving fragmented security practices

Desired State

- No longer reliant on ancient operating systems
- No more stagnant migration
- Secure systems with cybersecurity mindset

People, Process, Technology

Motion Consulting

Current State

- Using legacy Honeywell TDC 3000, which is over 40 years old
- Partial migration to Siemens PCS 7, leaving fragmented security practices

Key Enabler

Prairie Oil needs a new system that allows for critical processes to be protected

Desired State

- No longer reliant on ancient operating systems
- No more stagnant migration
- Secure systems with cybersecurity mindset

Key Takeaway

The new system must provide up-to-date cyber threat security measures

Criteria:
Cybersecurity

People, Process, Technology

Motion Consulting

Current State

- Using Honeywell TDC 3000 which in 2025 will be ... years old
- Partial migration to Siemens PCS 7, leaving fragmented security practices

Key Enabler

... that allows for critical processes to be protected

Desired State

- No longer reliant on ancient operating systems
- No more stagnant migration
- Secure systems with cybersecurity mindset

Stakeholder Analysis

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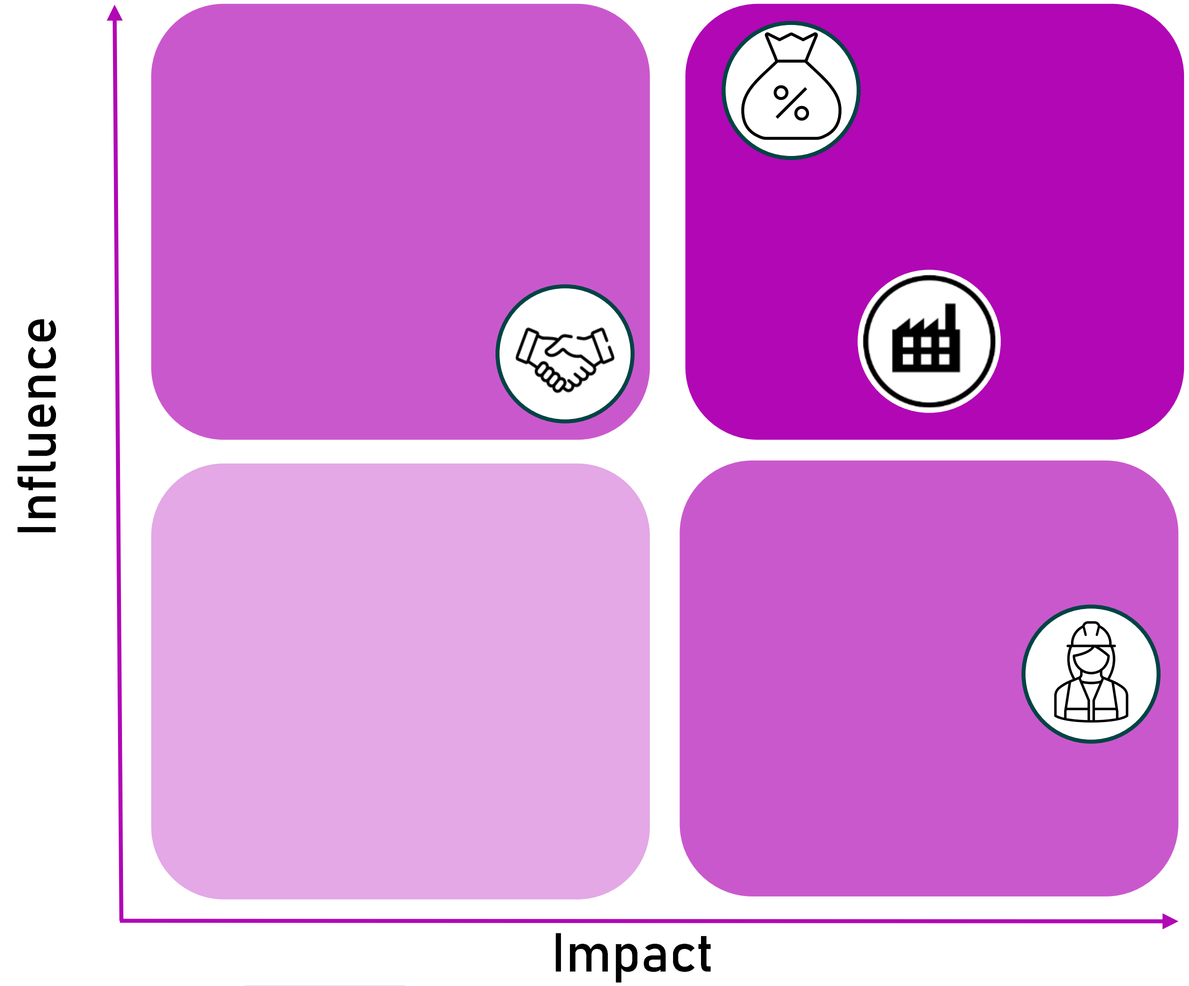


- Prairie Oil Executives
- Suppliers
- Distributors
- Prairie Oil Workforce

Stakeholder Analysis

Motion Consulting

Key Takeaway

The solution must be financially feasible to ensure all parties are not negatively affected

Criteria:
Financial Feasibility

Influence

Impact

Prairie Oil Executives

Suppliers

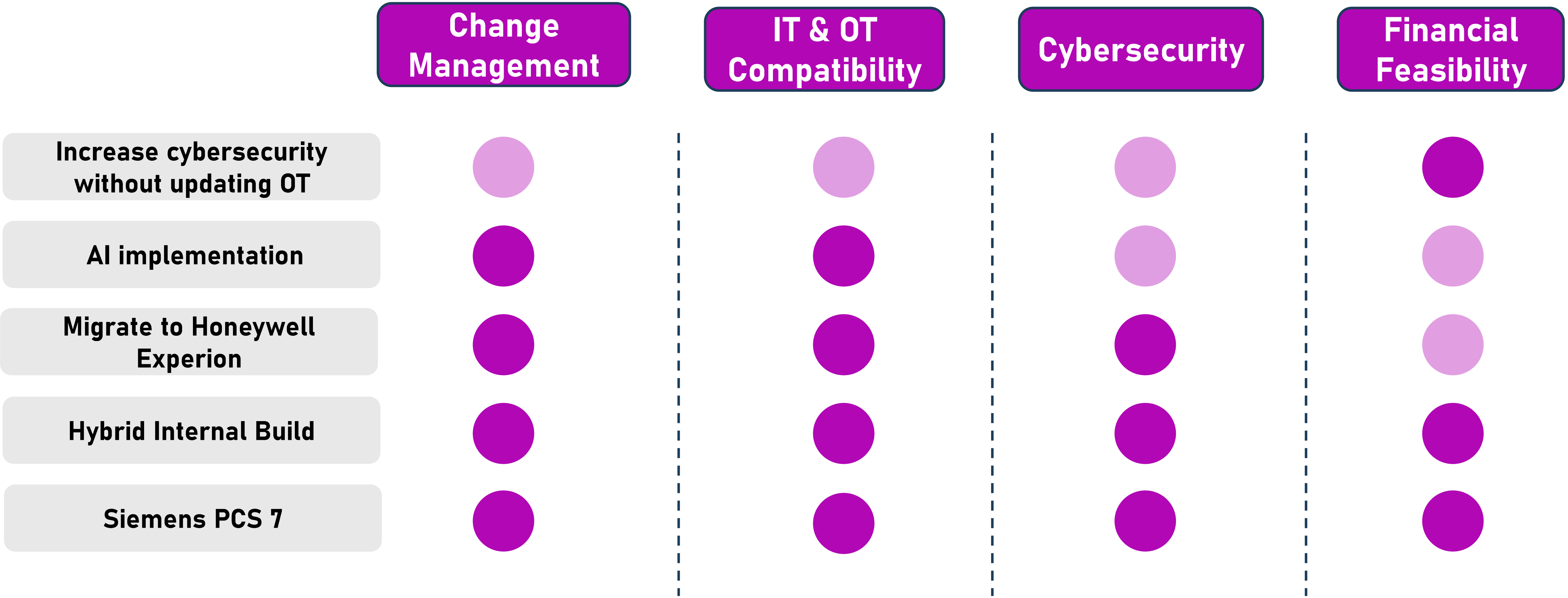
Distributors

Prairie Oil Workforce

Alternatives

Motion Consulting

Assessing potential strategies against criteria



Alternatives

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Assessing potential strategies against criteria

	Change Management	IT & OT Compatibility	Cybersecurity	Financial Feasibility
Increase cybersecurity without updating OT				
AI implementation				
Migrate to Honeywell Experion				
Hybrid Internal Build				
Siemens PCS 7				

Keep It Moving

1

Data Integration and Migration

- Move from siloed data into AWS
- Implement backup and encryption

2

Internal Structural Changes

- Ensure OT and IT compatibility
- Network segmentation

Keep It Moving

Goal:

Prioritize cyber safety with minimal downtime

1

Data Integration and Migration

- Move from siloed data into AWS
- Implement backup and encryption

2

Internal Structural Changes

- Ensure OT and IT compatibility
- Network segmentation

Data Integration and Migration

Phase 1

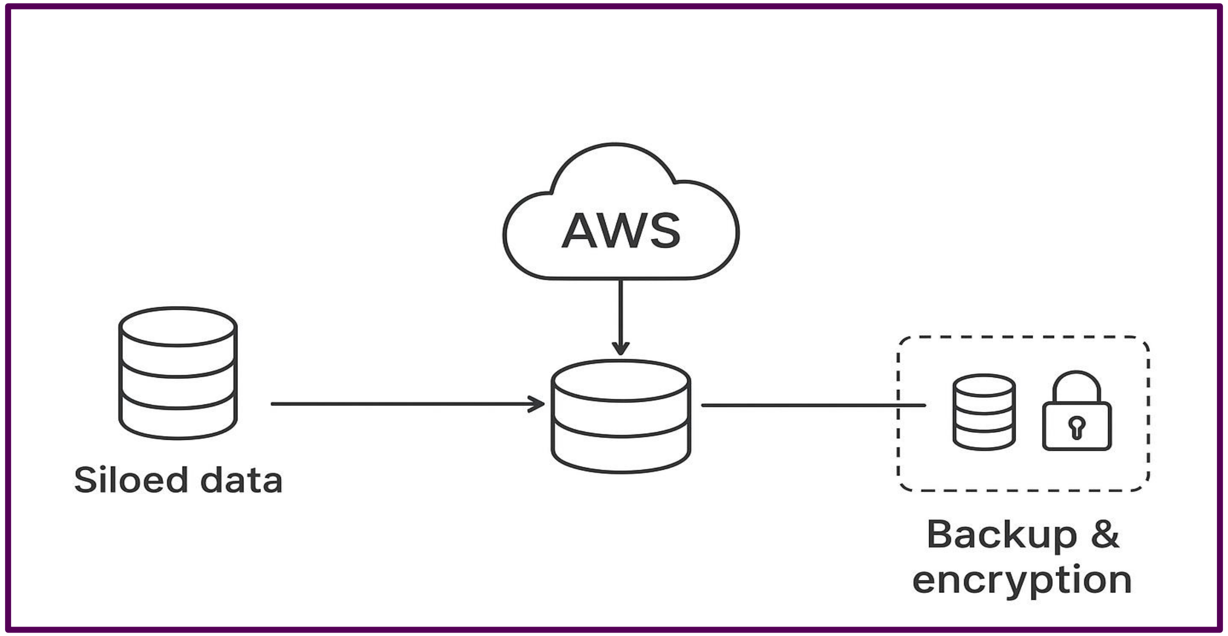
Transition to centralized data lake (AWS)

Key Features:

AWS Identity Access and Management:
User/role permissions for access

AWS Key Management Service:
Encryption Key Management

AWS Audit Manager:
Automates evidence collection for audits



Internal Structural Changes

Phase 2

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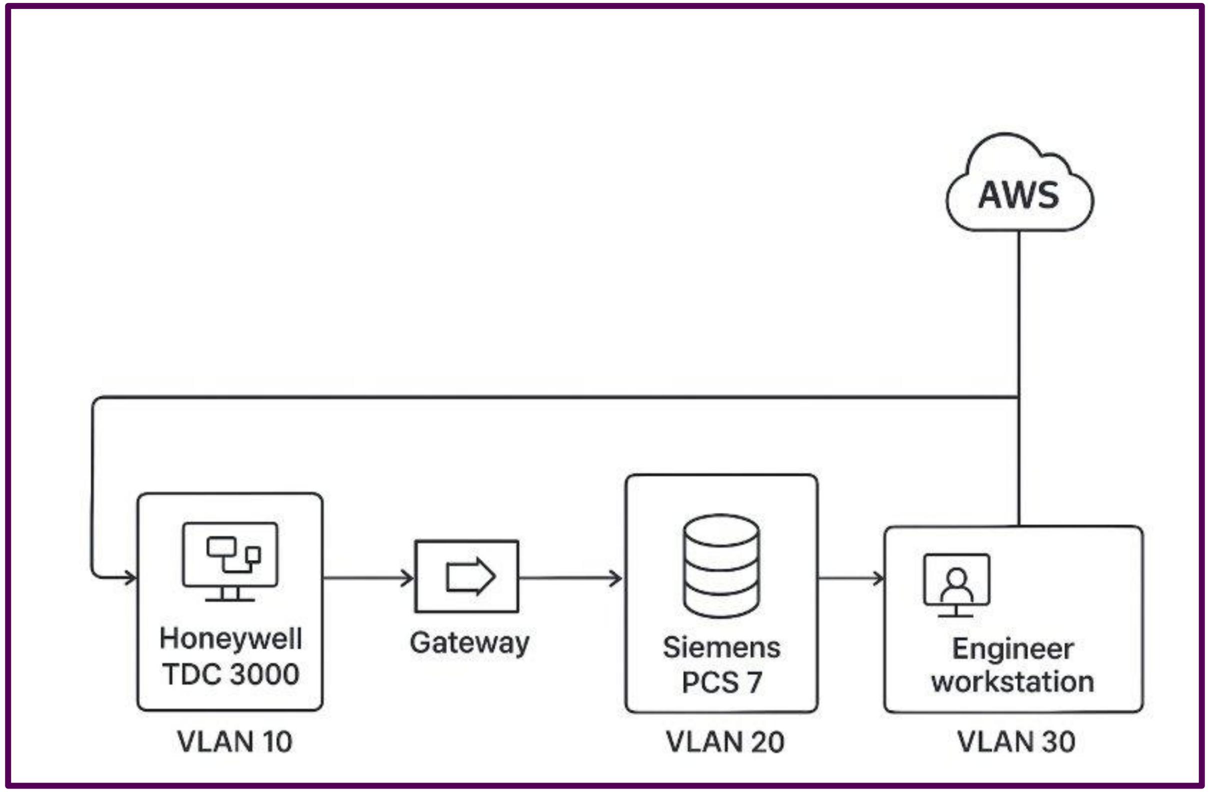


Migration to Siemens PCS 7

Key Features:

Integrated DCS Platform
Process control, safety systems, batch control, and asset management in one environment.

Scalability and Flexibility
Handles anything from a single compressor station up to enterprise-wide pipeline operations across thousands of I/O points.

Cybersecurity Standard
Built to align with IEC 62443 industrial cybersecurity standard.

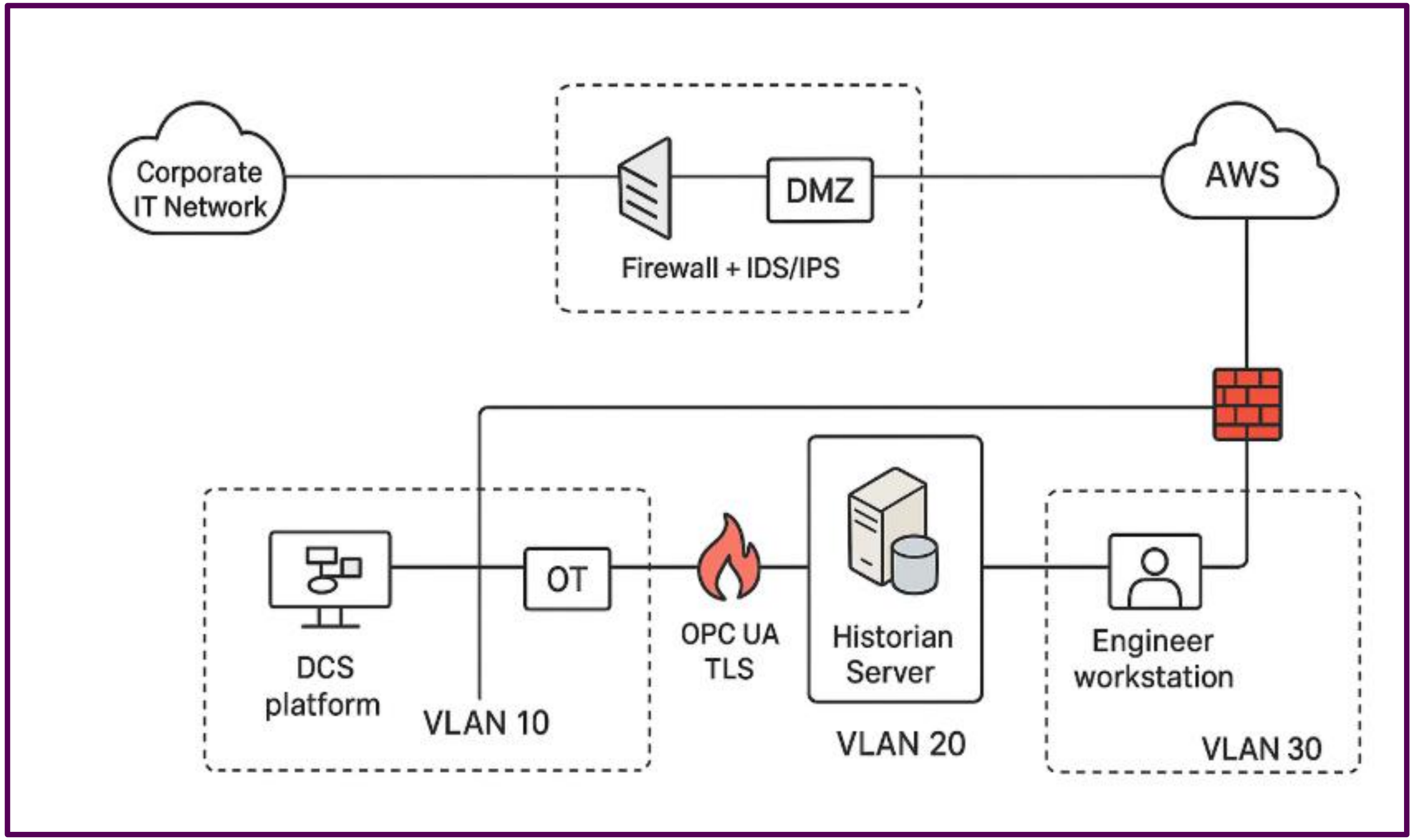
OT-IT Compatibility

Phase 3

Key Features:

Siemens PCS 7 (OT Backbone)
Role-based access, system hardening, patch management, secure communications, lifecycle support (15-20 years), and scalability.

AWS Security (IT Layer)
Cybersecurity and compliance, cloud-based replicas of control center HMIs for disaster recovery, extended visibility across both OT and IT.



Network Segmentation

Phase 4

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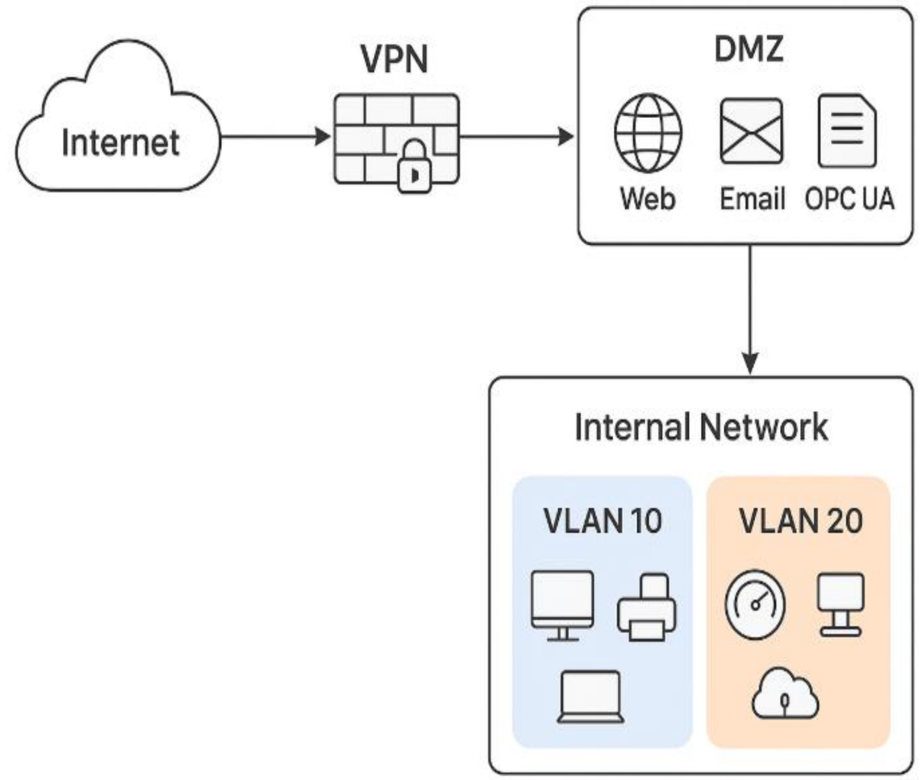

Enhance network security, control access, and protect critical systems

Key Features:

VPN Encrypted Remote Access:
Safe connection of external users to the internet.

DMZ Buffer Zone:
Hosts public-facing services while keeping internal systems protected.

VLAN Segmentation of Internal Network:
Limits lateral movement of threats.

Change Management - ADKAR

Ensuring everyone is involved.

Motion Consulting

Awareness

"The current system works fine, why do we need to change it?"

Growing security threats and aging equipment make this change necessary.

Is this possible??

Why change??

Vivek

Kaitlynn

Change Management - ADKAR

Ensuring everyone is involved.

Motion Consulting

Awareness	"The current system works fine, why do we need to change it?"	Growing security threats and aging equipment make this change necessary.
Desire	"Why should I have to learn how to use this new technology?"	The updated system will reduce workload and shutdown time.

Change Management - ADKAR

Ensuring everyone is involved.

Motion Consulting

Awareness	"The current system works fine, why do we need to change it?"	Growing security threats and aging equipment make this change necessary.
Desire	"Why should I have to learn how to use this new technology?"	The updated system will reduce workload and shutdown time.
Knowledge	"Is there some way I can get a refresher on the new system?"	Employees will be thoroughly trained in new system prior to operational launch.

Change Management - ADKAR

Ensuring everyone is involved.

Motion Consulting

Awareness	"The current system works fine, why do we need to change it?"	Growing security threats and aging equipment make this change necessary.
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Ability	"I feel like some of the MFA features are redundant."	Employees can provide feedback and solutions to the minor inconveniences.

Change Management - ADKAR

Ensuring everyone is involved.

Motion Consulting

Awareness	“The current system works fine, why do we need to change it?”	Growing security threats and aging equipment make this change necessary.
Desire	“Why should I have to learn how to use this new technology?”	The updated system will reduce workload and shutdown time.
Knowledge	“Is there some way I can get a refresher on the new system?”	Employees will be thoroughly trained in new system prior to operational launch.
Ability	“I feel like some of the MFA features are redundant.”	Employees can provide feedback and solutions to the minor inconveniences.
Reinforcement	“Will we ever go back to the old system?”	The new system is sustainable for Prairie Oil in the long-term.

Case Studies

Proof of previous implementation

RoviSys' oil and gas company IT/OT Integration

Challenged with ensuring data quality and normalization across nearly 100 facilities, the development of standard interfaces and data quality procedures was required to ensure rapid scale-out of the solution.

Migration of a SCADA system to IaaS clouds – A Case Study
Journal Of Cloud Computing

DTE Calvert City

Using gradual planning and an emphasis on reduced downtime, DTE Energy switched from the obsolete APACS DCS to a full Siemens PCS 7 system

Case Study

Moving the SCADA systems to an Infrastructure as a Service (IaaS) cloud allowed for: cheaper deployments, system redundancy support, and increased uptime.

Immediate Implementation

How can we get started?

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Tomorrow's Action Points

Kick off an organization-wide awareness campaign to show the need for change

Start AWS Cloud Transmission Process

Timeline

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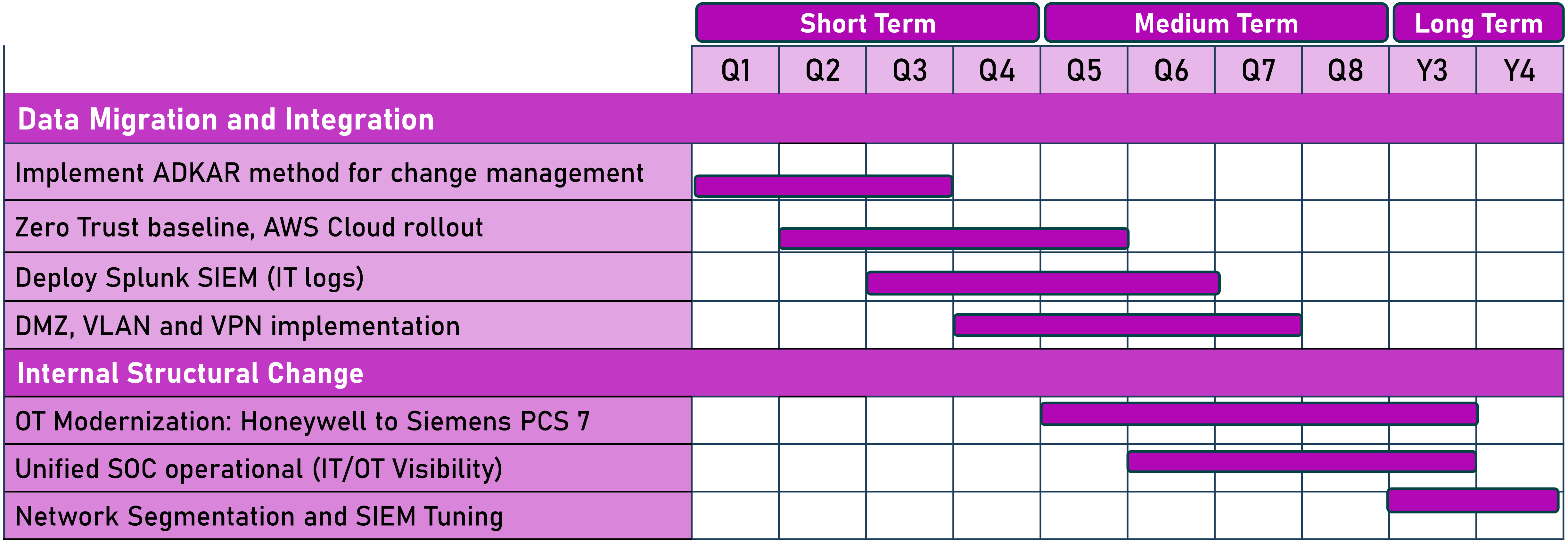
Consulting

Key tasks associated with the success of the implementation



Future Considerations

How can we keep it going?

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Year 5

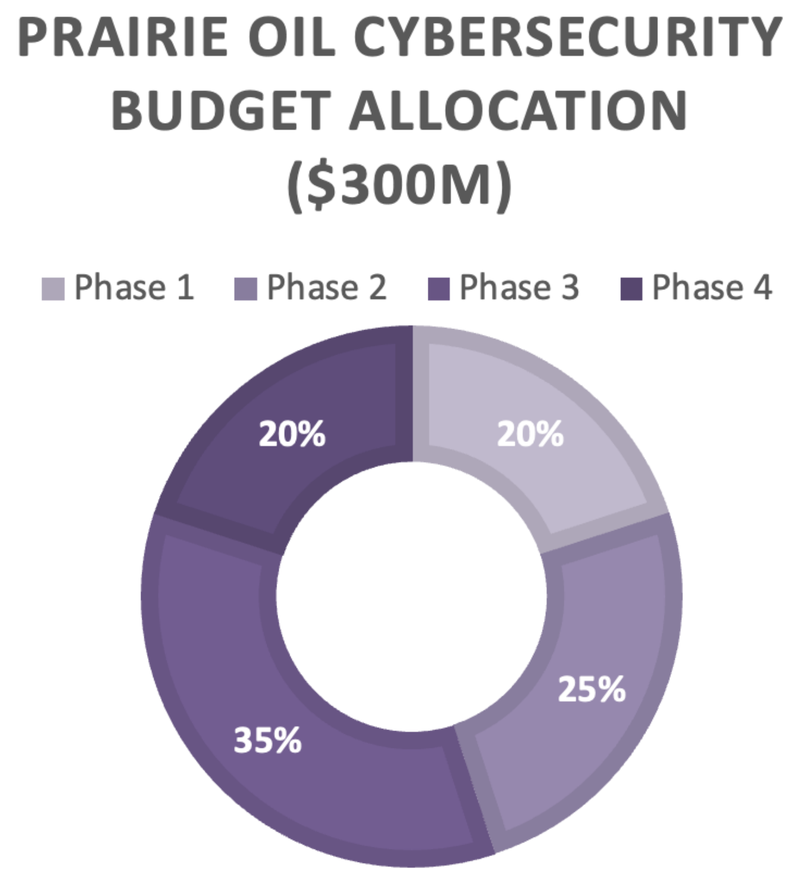
- Review and reassess AWS data storage
- Monitor organization adaptation to new processes

- Industry Benchmarking
- Consider AI/ML introduction

Financials

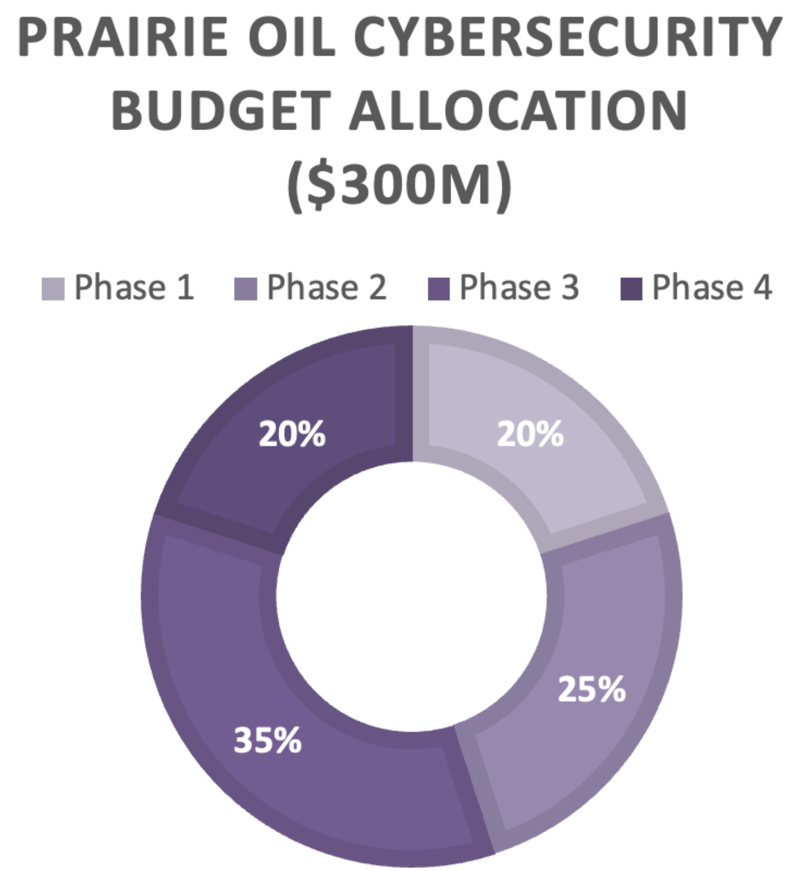
Motion Consulting

Phase 1	Cost (Million $)
ADKAR Change Implementation	15
Workforce Training & Certifications	10
Zero Trust Baseline MFA, AWS	20
Partner Compliance Audits	15
Phase 1 Total	**60**



Financials

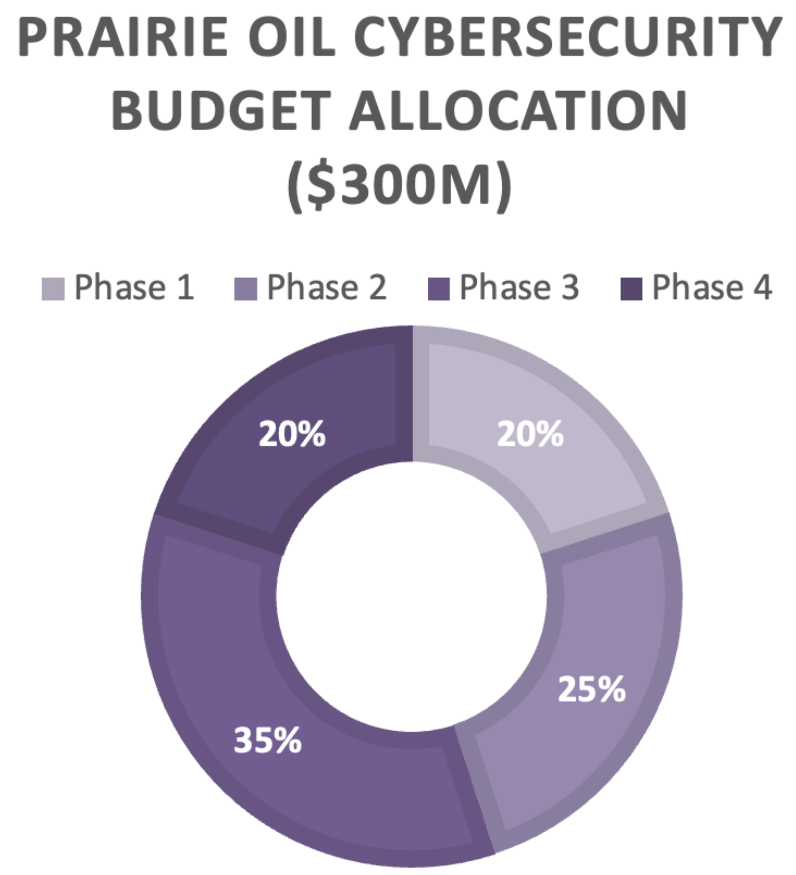
Phase 2	Cost (Million $)
Splunk SIEM Licenses & Cloud Hosting (AWS)	40
SOC Tools & Threat Intel Feeds	20
Integration & Deployment Services	15
Phase 2 Total	**75**



Financials

Motion Consulting

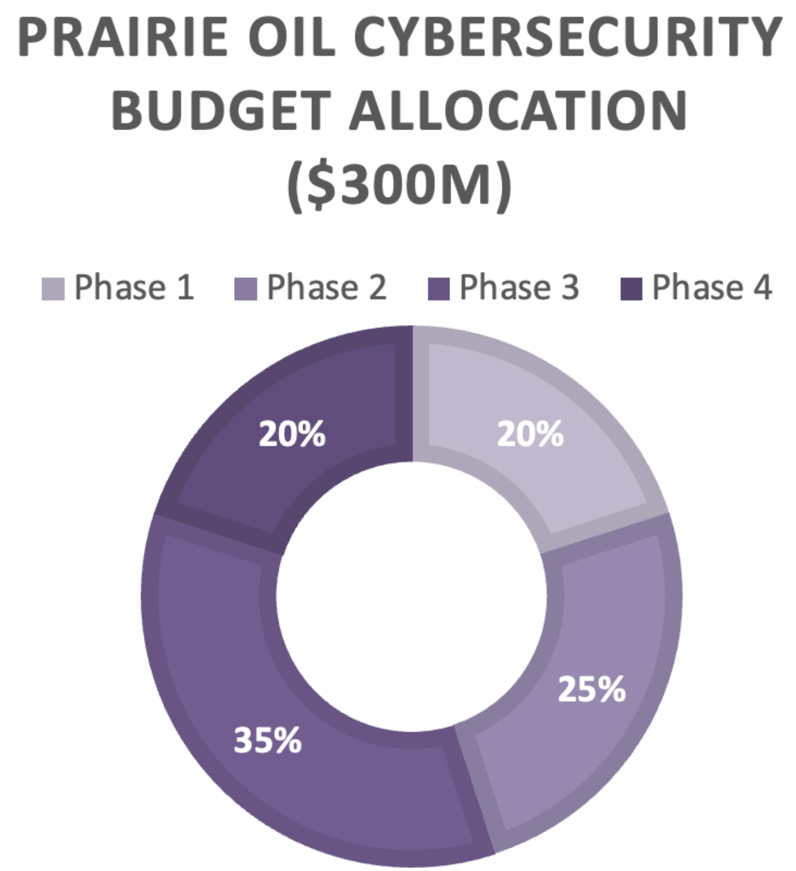
Phase 3	Cost (Million $)
Honeywell TDC 3000 → Siemens PCS 7 Migration	70
OT Firewalls, IDS, Secure Workstations	20
Network Segmentation & Controllers	15
Phase 3 Total	**105**



Financials

Motion Consulting

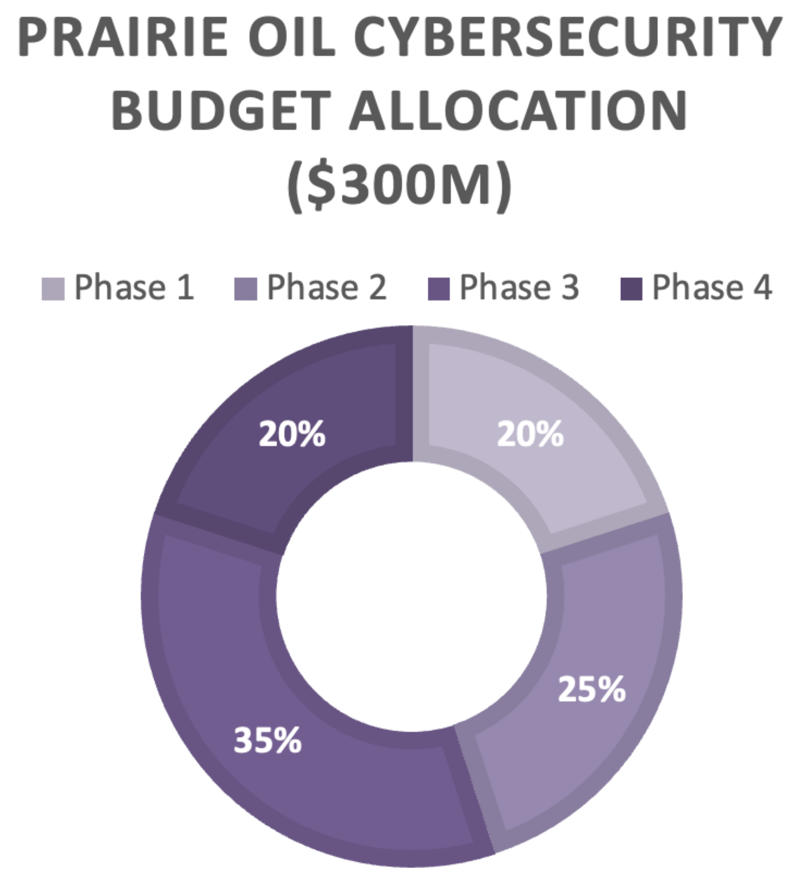
Phase 4	Cost (Million $)
Unified SOC Staffing & Operations	25
Immutable Backups & Disaster Recovery	15
Partner/Regulator Joint Exercises	10
Continuous Improvement & Refresh Training	10
Phase 4 Total	60



Financials

Motion Consulting

Phases	Cost (Million $)
Phase 1	60
Phase 2	75
Phase 3	105
Phase 4	60
Total	**300**



Key Performance Indicators

Assessing the performance of the implementation

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$4 Billion	24 Hours	Zero
Prairie Oil's investment prevents an OT shutdown, saving ~$4.0B in lost revenue.	Or less. Time needed to to detect and resolve Cybersecurity Incidents.	Zero major cyber incidents.

Personal Impacts

Back to Vivek & Kaitlynn

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Vivek

IT Lead

- Structured patch schedule that can align with plant operations, no more risk of downtime
- No longer dreads audits, knows exactly what his system is capable of because he was a part of the solution
- No more security tasks interfering with day-to-day IT duties

Kaitlynn

Pipeline Supervisor

- Lighter workload in unexpected ways, fewer nuisance alarms, intuitive technology
- Slowly regaining confidence in new operating systems
- Knew how to test, control and manage the system before it went up; her fingerprints were in the design

Risks and Mitigations

Proactive Control

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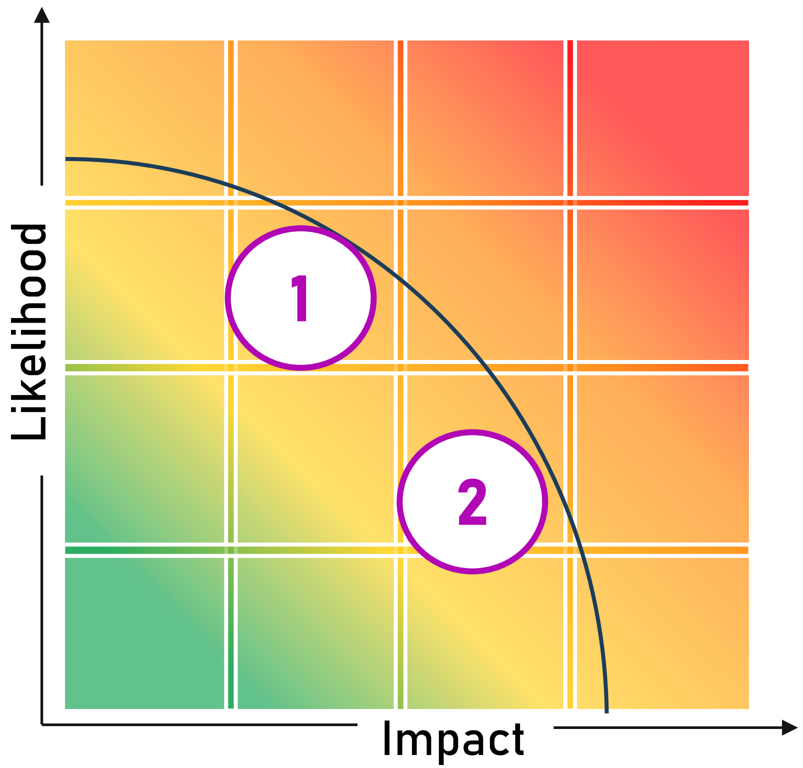
Risk Matrix



	Risk	Mitigation
1	**Resistance to Change**	Ensure employees are a part of the process, implementing the ADKAR method.
2	**Downtime During Transition**	Patching, segmentation, and system migrations done in phases. OT dual system during installation.

Executive Summary

How did we help?

Motion Consulting

Problem Statement	**How can Prairie Oil prepare for growing cybersecurity threats in the industry?**
Criteria	Change Management · IT & OT Compatibility · Cybersecurity · Financial Feasibility
Solution	***Keep It Moving***
Impacts	4 Billion Dollars · 24 Hours · Zero Incidents

Q&A

Prepared for

Prairie Oil

Diya Arora

Kritin Kumar

Daisy Dyck

Christian Arteaga

Motion Consulting

Appendix A

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Annual Revenue: $12.5B

Revenue at Risk (from a major cyberattack): ~$4.0B

Cybersecurity Investment: $0.3B (5 years)

Avoided Losses: ~$4.0B (Complete shutdown prevented, Shell Canada)

Revenue Growth Upside: ~$0.25B (contracts, uptime gains, trust benefits)